

Learn Python

Teacher Edition

Minute-by-minute session playbooks · transition scripts · predicted misconceptions · Socratic prompts

An 11-session, ~22-hour (two hours per session, + capstone) accelerated Python course for a PhD-in-Education learner with no prior coding experience.

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Learn Python — TEACHER Edition

Everything in the student syllabus, **plus** the instructor scaffolding: a minute-by-minute clock for each two-hour session, transition scripts (how to move between blocks without dead air), predicted misconceptions for *this* learner, Socratic prompts (DeepTutor-style — ask, don't tell), and an explicit "**if you're behind, cut this**" line per session.

Each two-hour session has two halves around a break: the **core** of the topic first, then a **Going deeper** block — genuinely new material (not stretched practice), with its own live demos and its own practice round. Sessions 1–10 run in five natural pairs (types→traps, control flow→data structures, functions→recursion, exceptions→files, regex→modules/OOP); Session 11 then revisits Session 8's data work with NumPy and pandas. Each pair shares a through-line; **name it out loud** when you open the second session of a pair, so consecutive sessions feel like one arc.

How to read this document

Each session has:

- **Covers** — core topic · going-deeper topics.
- **Pre-flight** — what to have open/ready before the student arrives.
- **The clock** — a 120-minute breakdown. Keep a timer visible. The numbers are targets, not law.
- **Transitions** — short scripted lines to hand off cleanly between blocks.
- **Predicted misconceptions** — where THIS learner (expert in research, novice in code) will stumble.
- **Socratic prompts** — questions to ask instead of explaining; let him derive it.
- **Cut line** — the first thing to drop if you're running over.
- **Homework** — what to assign at the close (specs live in `examples/session-NN/practice.md` → *Homework*).

For the exhaustive point-by-point tick-list of what each session must land (every concept, trap, practice task, and homework item, split core/deeper), print **Appendix A — the Session-by-Session Coverage Checklist** and keep it next to the clock.

The standard clock (adapt per session below)

- **0:00–0:05 — Warm-up.** Previous homework debrief + misconceptions log. Don't re-teach — quiz.
- **0:05–0:30 — Core concept.**
- **0:30–0:42 — Live demo I** (`demo.py` , predict-then-run; he types along).
- **0:42–1:00 — Practice I** (`practice.md` → *In class*).
- **1:00–1:08 — Break.** Honor it.
- **1:08–1:30 — Going deeper concept** (the deck's *Going deeper* slides).
- **1:30–1:40 — Live demo II** (the demo's *GOING DEEPER* sections).
- **1:40–1:53 — Practice II** (*In class — going deeper*).
- **1:53–2:00 — Recap + quiz + homework.**

Universal pacing principles (this learner is fast)

- **Talk less than you want to.** He reads fast and abstracts well. Default to "here's the rule, here's the trap, now you try."
 - **Protect both practice blocks** (~30 min total, plus the live coding he types along with). If concept overruns, steal from your own talking, never from his typing.
 - **The second hour is new material, not overflow.** If Practice I runs long, cut tasks, don't eat the Going-deeper block.
 - **Predict-then-run is the engine, especially the Session 2 traps.** Always have him *commit to an answer out loud* before running.
 - **The warm-up is homework-powered.** If homework wasn't done, do the first homework task together as the warm-up and move on.
 - **Don't pad.** If a block ends early, bank the minutes for practice.
 - **Carry a running "misconceptions log"** (DeepTutor "learning memory"): note every trap he hit; re-surface it as the next warm-up.
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SESSION 1 — Running Python, Variables & Types

Covers: REPL vs script, variables as labels, the five core types, `input()` / `print()`, f-strings, casting, tracebacks · **deeper:** the full operator set (`// % **`), string methods, aligned f-strings, `import math`, `help()` / `dir()`, naming. **Pre-flight:** terminal + VS Code open; `examples/session-01/` ready; a deliberately broken line staged for the traceback.

The clock (120 min)

- **0:00–0:05 — Orientation.** Why Python, why this re-ordered path, how a session runs (core → break → deeper), what homework is for.
- **0:05–0:30 — Core concept.** REPL vs script. Variables as *labels on objects* (Connection Map #1). The five core types; `type(x)`. `input()` → always `str`. f-strings. Casting; `int(3.9)` truncates.
- **0:30–0:42 — Live I.** Build `greet.py`, then "years to graduation"; trigger and *read* one traceback (last line first).
- **0:42–1:00 — Practice I.** GPA reporter, age bucket, the string-concatenation trap.
- **1:00–1:08 — Break.**
- **1:08–1:30 — Deeper concept.** Arithmetic fully (`/ // % **`, precedence, `+=`); `%`-as-remainder uses; string methods (chaining, immutability, repetition); alignment f-strings; `print(sep=, end=)` and escape sequences; `import math` ("borrowing a toolbox — full story in S8"); `help()` / `dir()`; naming conventions & constants.
- **1:30–1:40 — Live II.** The demo's deeper sections: minutes→h/min, `% 3` groups, `.strip().title()` chain, the aligned report.
- **1:40–1:53 — Practice II.** Aligned gradebook lines; name normalizer; whole-units-and-remainder.
- **1:53–2:00 — Recap + quiz + homework.** `input()` → string; `int()` truncates; methods return new strings. Quiz S1.

Transitions

- Concept→live: *"Enough theory — watch me make these mistakes so you don't have to."*
- Break→deeper: *"You can run code. Now let's make the code worth running — the operators and string tools you'll touch every single day."*
- Close: *"You know the five types. Next session is the whole reason this course exists: the ways Python lets those types surprise you."*

Predicted misconceptions

- Expects `input()` to give a number → `"5" + "3" == "53"`.
- Thinks `=` asserts equality (math habit). Label-on-object now; pays off in S2 and S4.
- Expects `int(3.9)` to round; expects `.upper()` to change the string in place.

Socratic prompts

- "What type do you think `input()` hands back? How could we check?"
- "130 minutes. How do you get 2 and 10 out of it with two operators?"

- "You cleaned the name but nothing changed. What did `.strip()` actually hand you?"

Cut line: drop `help()` / `dir()` and the naming slide (point at the cheat sheet); never cut the traceback reading or either practice block. **Homework:** unit converter · traceback drill · type-prediction table.

SESSION 2 — The Dynamic-Typing Traps (*the most important session*)

Covers: `==` vs `is`, `bool < int`, float precision, cross-type comparison, truthiness, `isinstance` vs `type` · **deeper:** `None` semantics, the conversion matrix, `nan / inf`, mutable vs immutable + `id()`, `Decimal`. **Pre-flight:** `examples/session-02/`; the trap gauntlet open; `cheatsheets/traps-and-gotchas.md` ready to hand over.

The clock (120 min)

- **0:00–0:05 — Warm-up.** S1 homework: one traceback from his drill, one type-table row.
- **0:05–0:25 — Core concept.** This is the heart of the course; tell him so. `==` value vs `is` identity (Connection Map #3), `is None`; `bool < int` (Connection Map #4); floats are binary → `math.isclose` (Connection Map #5); `5 == "5"` vs `5 > "5"`; `isinstance` vs `type`; truthiness.
- **0:25–0:45 — The gauntlet (live + practice fused).** ~18 one-line traps, his prediction *out loud* before every run.
- **0:45–1:00 — Practice I.** The paper gauntlet, then start `clean_score()`.
- **1:00–1:08 — Break.**
- **1:08–1:30 — Deeper concept.** `None` properly (three kinds of "nothing"); the conversion matrix ("convert at the edge, once"); `nan / inf` and `math.isnan`; mutable vs immutable table + `id()` (*this explains every is surprise you just saw*); `Decimal` for exact arithmetic — and why you build it from strings.
- **1:30–1:40 — Live II.** Demo deeper sections: conversion matrix, `nan == nan`, `id()` on alias vs equal, `Decimal("0.1") + Decimal("0.2")`.
- **1:40–1:53 — Practice II.** Conversion predictions; the `Decimal` re-run; `describe(x)` (`None` vs `""` vs `0` vs `False` — his first real use of `isinstance(bool)` ordering).
- **1:53–2:00 — Recap + quiz + homework.** Hand over the trap cheat sheet as his permanent reference. Quiz S2.

Transitions

- Warm-up→concept: *"You know the five types. Here's the catch: Python lets them mix in ways that surprise everyone."*
- Break→deeper: *"You've seen WHAT surprises. Now the machinery underneath — what None really is, what id() shows, and the tool for when floats aren't good enough."*
- Close: *"Every trap you predicted wrong goes in your journal tonight — that's the homework, and it becomes your personal cheat sheet."*

Predicted misconceptions

- Assumes `is` is a stylistic `==` — nail it with the mutable-list demo, then `id()`.
- Trusts `==` on floats ("I round anyway") — the *cause* is binary storage, not data.
- Will write `Decimal(0.1)` and wonder why nothing improved — strings, not floats.
- `describe(False)`: forgets `False == 0` and reports "zero" — that's the planned stumble.

Socratic prompts

- "Two students, same 3.7 GPA. `==` ? `is` ? Why?"
- " `0.1 + 0.2` isn't `0.3` — is the bug in the data or in the computer? And when is `isclose` not good enough?" (→ money/bookkeeping → `Decimal`)
- "What's the difference between a blank cell, a zero score, and a missing student?"

Cut line: drop `inf`, `Fraction`, and the small-int cache wrinkle; **never** cut `==` / `is`, floats, `isinstance`, the gauntlet, or `describe(x)`. **Homework:** trap journal · `approx_equal` · `is_missing`.

SESSION 3 — Control Flow: Conditionals & Loops

Covers: `if/elif/else`, chained comparisons, `and/or/not`, `for/while`, `range`, `break/continue`, `enumerate/zip`, the validation loop · **deeper:** the ternary, `match/case`, `for/else`, nested loops, the named loop patterns. **Pre-flight:** `examples/session-03/`; a messy nested-`if` staged for refactor; `Ctrl+C` ready for the infinite loop.

The clock (120 min)

- **0:00–0:05 — Warm-up.** S2 homework: two trap-journal entries; quiz `approx_equal(0.1+0.2, 0.3)`.
- **0:05–0:30 — Core concept.** `if/elif/else` — and `if` vs `elif` (stacked `if`s are independent questions that ALL fire; show the double-label bug); **chained comparisons**; `and/or/not`, short-circuit, operand-return; `while` (+ infinite-loop demo); `for ... in`; `range` off-by-one; `break/continue`; the `while True`: validation pattern; `enumerate/zip` vs `range(len(...))`.
- **0:30–0:42 — Live I.** Likert→label classifier; roster average two ways; the "ask until valid" loop.
- **0:42–1:00 — Practice I.** Grade-band classifier (**test every boundary**: 89.999/90/90.001), logic drill, `zip` pass/fail, validation loop, the mutate-while-iterating trap.
- **1:00–1:08 — Break.**
- **1:08–1:30 — Deeper concept.** The ternary (tiny choices only); the boolean-return idiom (`return score >= 60` — hand the comparison straight back, use it bare in `if`); `match/case` (same-value-vs-literals ladders); `for/else` (search without a flag); nested loops (`break` exits inner only → function-and-`return` escape); the named patterns: accumulator, counter, best-so-far, sentinel.
- **1:30–1:40 — Live II.** Demo deeper sections: `match/case` classifier, `for/else` first-failing, nested sections×students, best-so-far.
- **1:40–1:53 — Practice II.** `match/case` rewrite (then: which reads better, and why?); `for/else` search; best-and-worst in one pass; the return-the-test rewrite.
- **1:53–2:00 — Recap + quiz + homework.** `if x == True` → `if x`; `range` excludes stop; name your pattern before you type. Quiz S3.

Transitions

- Concept→live: *"Watch me turn an ugly five-level `if` into three readable lines."*
- Break→deeper: *"You can branch and repeat. Now the idioms that make control flow READ well — and the four skeletons behind almost every loop you'll ever write."*
- Close: *"Next session: the containers worth looping over — and a list of dicts is secretly your dataset."*

Predicted misconceptions

- Writes `if score >= 90 and score < 100` — chain it.

- Boundary errors at cutoffs — make him test 89.999/90/90.001.
- `range(len(x))` habit — push `enumerate / zip`.
- Will over-apply `match/case` everywhere for a week — name its niche (one value vs literals).

Socratic prompts

- "`x = 5` and `0` — what's `x`? Why isn't it `True / False`?"
- "When does the `else` on a `for` run? What bookkeeping does it delete?"
- "Which of the four patterns is your grade-average loop? Your validation loop?"
- "A 95 printed three labels. What question did each `if` actually ask?"

Cut line: drop nested loops and the ternary slide (they resurface naturally); keep `match/case`, `for/else`, and both practice blocks. **Homework:** attendance labeler · number-guessing game · leap-year checker.

SESSION 4 — Data Structures

Covers: `list` / `tuple` / `dict` / `set` , slicing, list-of-dicts-as-dataset, comprehensions, `sorted(key=...)` , aliasing · **deeper:** unpacking, dict power methods, `Counter` / `defaultdict` , full set algebra, multi-key sorting, `copy` vs `deepcopy` . **Pre-flight:** `examples/session-04/` ; the "list of dicts = tidy dataset" diagram (Connection Map #6); the aliasing demo staged.

The clock (120 min)

- **0:00–0:05 — Warm-up.** S3 homework: `label(79.9) ? is_leap(1900) ?` Re-run one boundary he missed.
- **0:05–0:30 — Core concept.** The four containers; indexing & slicing; nesting → **a list of dicts is a dataset**; list/dict comprehensions; `sorted(key=lambda ...)` .
- **0:30–0:42 — Live I.** Build the roster, sort by score, dedupe with a `set` , rewrite a loop as a comprehension — then the **aliasing** bug and its fix.
- **0:42–1:00 — Practice I.** Rank, `{name: score}` comprehension, group pass/fail, dedupe, reproduce-then-fix aliasing.
- **1:00–1:08 — Break.**
- **1:08–1:30 — Deeper concept.** Unpacking (`head, *rest` , zip-loops ARE unpacking); dict power methods (`.get/.setdefault/.pop` , `| merge`); `Counter` ("your tally loop, one line"); `defaultdict(list)` grouping; set algebra (`& | - ^` , subset, `dict.fromkeys` ordered dedupe); multi-key tuple sorts + stability; `copy` vs `deepcopy` .
- **1:30–1:40 — Live II.** Demo deeper sections: shallow-vs-deep copy on nested lists, ordered dedupe; plus `Counter/defaultdict` from the earlier sections if not yet run.
- **1:40–1:53 — Practice II.** `Counter` + `.most_common` ; `defaultdict` grouping; the multi-key sort; ordered dedupe.
- **1:53–2:00 — Recap + quiz + homework.** `=` aliases; `.sort()` returns `None` ; tuple keys sort on multiple fields; nested → `deepcopy` . Quiz S4.

Transitions

- Open (the pair's thread): *"You can loop over data. This session: the containers worth looping — and a list of dicts is just a tidy dataset."*
- Break→deeper: *"You can store and sort. Now the toolkit that deletes half your loops — Counter, defaultdict, and sorts with tie-breaks."*
- Close: *"Data's handled. Next session we stop repeating ourselves: functions."*

Predicted misconceptions

- **Aliasing** is the big one — `b = a` doesn't copy; tie to S1 "label on object" and S2 `id()` .
- Expects `xs.sort()` to return the list.
- Expects `set` to keep first-seen order — show `dict.fromkeys` .
- Multi-key sort: will try `reverse=True` when only ONE key should descend — show the `-score` trick.

Socratic prompts

- `" b = a; a.append(99)` — what's in `b` ? Why?"
- "Your tally loop is six lines. What one import makes it one line?"
- "Sort by score descending, ties by name ascending — what must the key return?"

Cut line: drop `^` symmetric difference and `deepcopy` detail (homework's grid covers sharing); keep Counter, defaultdict, and multi-key sorting. **Homework:** gradebook dict drill · frequency counter · fix-the-grid.

SESSION 5 — Functions, Scope & Reusability

Covers: `def`, parameters (positional/keyword/default, `*args` / `**kwargs`), `return` vs `print`, LEGB, docstrings, type hints, the mutable-default bug · **deeper:** functions as values, `lambda`, closures, a first decorator, keyword-only args, doctests. **Pre-flight:** `examples/session-05/`; the mutable-default demo staged (the headline); S4 inline code ready to refactor.

The clock (120 min)

- **0:00–0:05 — Warm-up.** S4 homework: the grid bug — *why* did every row change? (Aliasing. It should be automatic by now.)
- **0:05–0:30 — Core concept.** `def`; parameters; `return` vs `print`; scope (LEGB), why not `global`; docstrings; type hints (not enforced).
- **0:30–0:42 — Live I.** Refactor S4 code into `class_average` / `letter_grade`; the **mutable-default bug** live → fix with `None`.
- **0:42–1:00 — Practice I.** Grade-functions library; reproduce-then-fix the mutable default; `summary(*scores)`.
- **1:00–1:08 — Break.**
- **1:08–1:30 — Deeper concept.** Functions are objects (assign, pass, dispatch dict) — `key=` was *never magic*; `lambda` honestly; closures (`make_curver` factory; the capture-by-variable trap); a first decorator (`@announce` — "you'll USE `@property`, `@cache`, `@pytest.mark` forever; now you know what `@` does"); keyword-only `*`; doctests.
- **1:30–1:40 — Live II.** Demo deeper sections: the stats dispatch dict, curver factory, doctest run.
- **1:40–1:53 — Practice II.** `apply_to_all`; `make_curver`; write `@announce`; add a doctest and run it.
- **1:53–2:00 — Recap + quiz + homework.** Mutable default → `None`; forgot `return` → `None`; a decorator wraps; doctests keep docs honest. Quiz S5.

Transitions

- Concept→live: *"This next bug has burned every Python programmer once. Watch the list keep growing."*
- Break→deeper: *"You can write functions. Second hour: functions AS VALUES — the idea behind `key=`, decorators, and half of modern Python."*
- Close: *"A function can call other functions. Next session, the mind-bender: it can call itself."*

Predicted misconceptions

- Won't believe the default list persists — show twice; defaults evaluate once, at `def`.
- Thinks printing is returning.
- Decorators look like magic — write `curve = announce(curve)` long-hand FIRST, then show `@` is only sugar.

Socratic prompts

- "When does `bag=[]` actually run?"

- "What exactly did `sorted` receive when you wrote `key=lambda s: s['score']` last session?"
- "`make_curver(5)` returned something. What is it? Where does the 5 live now?"

Cut line: drop keyword-only args and `nonlocal`; **never** cut the mutable-default demo or the decorator (S6's `@cache` and S10's `@property` depend on it). **Homework:** tiny stats library · `mean_ignoring_none` · scope prediction.

SESSION 6 — Recursion & Recursive Thinking

Covers: base + recursive case, the call stack, recursion vs iteration, nested data, `RecursionError` · **deeper:** memoization (`@cache`), binary search, tree data, the explicit-stack escape, the decision checklist. **Pre-flight:** `examples/session-06/` ; nested dict staged for `deep_sum` ; the `runaway` overflow queued.

The clock (120 min)

- **0:00–0:05 — Warm-up.** S5 homework: `mean_ignoring_none(None)` ? The scope drill — what error, why?
- **0:05–0:30 — Core concept.** Base case + recursive case; trace `orderings(3)` as a stack; recursion vs iteration; the cost (frames, no tail-call optimization, limit ≈ 1000); the prereq-chain example.
- **0:30–0:42 — Live I.** `countdown` / `factorial` ; `deep_sum` on nested data; a `RecursionError` read together.
- **0:42–1:00 — Practice I.** Recursive `total` , `reverse` , `flatten` , `depth` , the two trap-fixes.
- **1:00–1:08 — Break.**
- **1:08–1:30 — Deeper concept.** Memoization: naive `fib` re-solves subproblems → `@functools.cache` (a decorator — yesterday's lesson working today); divide & conquer: binary search halves the problem ($\log_2$ intuition: $1,000 \rightarrow \sim 10$ steps); tree-shaped data (org chart: the function mirrors the tree); the explicit-stack version of `flatten` (same logic, no frames); the checklist: flat→loop, nested→recursion, repeated→cache, deep→stack.
- **1:30–1:40 — Live II.** Demo deeper sections: binary search with indented printout, org-tree count, `flatten_iter` .
- **1:40–1:53 — Practice II.** Binary search with step counter; `count_people` + `deepest` ; `flatten` without recursion.
- **1:53–2:00 — Recap + quiz + homework.** Reachable base case; `return` the call; pick by the shape of the data. Quiz S6.

Transitions

- Open (the pair's thread): *"A function can call other functions. The mind-bender: it can call itself — exactly the tool for data defined in terms of itself."*
- Break→deeper: *"You can write A recursion. Now: when it's brilliant (halving, trees, caching) and how to escape when it isn't (the explicit stack)."*
- Close: *"You handle clean data beautifully. Next session: what to do when the data fights back."*

Predicted misconceptions

- Forgets to `return` the recursive call → silent `None` .
- Thinks binary search works on unsorted data — ask him why sorted matters.
- Assumes `@cache` helps every recursion — only *repeated* subproblems (`deep_sum` gains nothing).

Socratic prompts

- "What's the smallest input where the answer is obvious? That's your base case."
- "fib(35) computes fib(5) how many times? What would 'remembering' buy?"
- "1,000 sorted names — how many looks to find one? Why?"

Cut line: drop `deepest` and the mutual-recursion aside; **never** cut `deep_sum`, `@cache`, or binary search. **Homework:** `sum_digits` · `deep_count` · loop-vs-recursion paragraph.

SESSION 7 — Exceptions & Defensive Code

Covers: `try / except / else / finally`, exception types, `raise`, EAFP vs LBYL, `assert`, a first pytest test · **deeper:** the exception hierarchy & handler order, custom exceptions with data, `raise ... from`, logging, parametrized tests. **Pre-flight:** `examples/session-07/`; the dirty list staged; `pytest` installed and verified.

The clock (120 min)

- **0:00–0:05 — Warm-up.** S6 homework: trace `sum_digits(4823)`; why did `deep_count` need the bool check? (S2 pays rent.)
- **0:05–0:30 — Core concept.** Errors vs exceptions; `try/except/else/finally` — and keep the `try` block minimal; common types on sight; `raise`; EAFP vs LBYL; `assert` (developer check, not validation).
- **0:30–0:42 — Live I.** Harden `safe_int() / clean_likert()`; a first `pytest.raises` test, run it green.
- **0:42–1:00 — Practice I.** Clean the raw list into values + a rejection log; add a `raise`; one pytest test.
- **1:00–1:08 — Break.**
- **1:08–1:30 — Deeper concept.** The exception family tree; handlers tried top-down → specific before broad; catching a tuple; custom exceptions that carry data (`.value`); `raise ... from` (domain error outside, real cause inside); `finally` → the cleanup mindset (`with` is this, packaged — S8 preview); `logging` vs `print` (output vs diary, levels); pytest round 2: parametrize, arrange-act-assert, test the edges.
- **1:30–1:40 — Live II.** Demo deeper sections: except-order, the logging rejection log.
- **1:40–1:53 — Practice II.** Fix the except-order bug; `SurveyError` with `raise ... from`; convert the log to `logging`; parametrize the tests.
- **1:53–2:00 — Recap + quiz + homework.** Specific before broad; never bare `except:`; log the diary, print the output. Quiz S7.

Transitions

- Open: *"Your real survey data WILL have 'N/A' in a numeric column. Today we write code that shrugs it off."*
- Break→deeper: *"You can catch an error. Now: catch the RIGHT error, keep the evidence, and leave a paper trail — that's the difference between a script and a tool."*
- Close: *"You can survive one bad value. Real data is a file full of them — next session we open a real CSV."*

Predicted misconceptions

- Writes bare `except:` or puts `except Exception` first — the unreachable-handler demo cures it.
- Confuses `raise` with `return`, `assert` with validation.
- Will log at `DEBUG` and wonder where the output went — show the level threshold.

Socratic prompts

- "'seven' instead of 7 — catch it before (`if`) or after (`try`)? Trade-offs?"
- "Your program rejected four values silently. Who finds out, and when?"
- "Why does the order of the two `except` clauses matter? What's tried first?"

Cut line: drop `raise ... from` and shrink logging to the two-line teaser; keep except-order, `SurveyError`, and parametrize. **Homework:** `ask_int` · three pytest cases (as one parametrized test if he's ahead) · error triage.

SESSION 8 — Files, Libraries & Research Data

Covers: `open / with`, file modes, CSV via `DictReader / DictWriter`, the researcher's stdlib, the pandas teaser · **deeper:** `pathlib`, encodings, `csv.reader` vs `DictReader`, JSON for nested data, `datetime`, seeded `random`. **Pre-flight:** `examples/session-08/` with `students.csv` + `survey.csv`; `pip` works; `pandas` importable.

The clock (120 min)

- **0:00–0:05 — Warm-up.** S7 homework: one error-triage item; `ask_int(" 42 ")` — why does EAFP win?
- **0:05–0:30 — Core concept.** `open / with`; modes (`"w"` truncates!); CSV as dicts (ties to S4); the researcher's stdlib (`statistics`, `pathlib` first taste); `pip install`.
- **0:30–0:42 — Live I.** Read `students.csv` → list of dicts, class mean, write a summary CSV.
- **0:42–1:00 — Practice I.** Per-item survey means skipping dirty values; write `survey_summary.csv`; mean by major.
- **1:00–1:08 — Break.**
- **1:08–1:30 — Deeper concept.** `pathlib` full tour (`/` joins, `glob / rglob`, `read_text`); encodings (`utf-8`, the Excel BOM → `utf-8-sig`, mojibake); `csv.reader` vs `DictReader`, `delimiter=","`; JSON for nested data (`True` → `true`, `None` → `null`); `datetime` (`strptime / strftime`, date math); `random` with a **seed** (reproducible sampling — methods-section gold); scripts that take arguments (`sys.argv`, `sys.exit`, the `argparse` pointer); `pip` & `PyPI` via `cowsay` (the point in one moo: installing is trivial); the `requests` API taste; binary files → a Pillow GIF; the extended pandas teaser.
- **1:30–1:40 — Live II.** Demo deeper sections: `csv.reader` rows, date math, the seeded sample run twice.
- **1:40–1:53 — Practice II.** `pathlib` inventory; days-between; seeded sample; CSV → JSON; the `argv`-guarded `report.py`; the Pillow GIF.
- **1:53–2:00 — Recap + quiz + homework.** `"w"` destroys; CSV values are strings; seed your randomness; JSON is its own language. Quiz S8.

Transitions

- Open (the pair's thread): *"You can survive one bad value. Real data is a file full of them."*
- Break→deeper: *"You've done one CSV by hand. Now the ecosystem around it — paths, encodings, dates, JSON, and reproducible sampling. This is the session your future scripts live in."*
- Teaser framing: *"Everything you did by hand, pandas does in five lines — next course, not today. Now you know what it does underneath."*
- Close: *"Numbers are clean. Next session: the messiest data of all — text."*

Predicted misconceptions

- Opens with `"w"` to read; iterates a file twice.
- Forgets CSV values are strings (`"91" + 1` — S2 again).
- Believes seeding makes randomness "fake" — reframe: it makes the *method* reproducible.

- Expects JSON to accept `True / None` spellings.

Socratic prompts

- "Why does `with` matter even if the program crashes?"
- "Your co-author reruns the sampling script. What guarantees they get the same participants?"
- "Nested per-item stats — why does CSV fight you and JSON doesn't?"

Cut line: drop cowsay, the Pillow GIF, encodings, and `csv.reader` (leave the slides as reference); shrink the pandas/API teasers; keep pathlib, datetime, argv scripts, and the seeded sample.

Homework: attendance-report pipeline · JSON round-trip · your own CSV.

SESSION 9 — Regular Expressions & Text Cleaning

Covers: raw strings, the survival tokens, `search / fullmatch / findall / sub`, capture groups, when NOT to use regex · **deeper:** flags, `re.compile`, `VERBOSE`, named/non-capturing groups, greedy vs lazy, `sub` with a function, regex + files. **Pre-flight:** `examples/session-09/`; messy text, emails, "Last, First" names staged; `regex101.com` open (flavor: Python); a whiteboard for state-machine sketches — drawing a pattern as a chain of hops makes greedy vs lazy visible.

The clock (120 min)

- **0:00–0:05 — Warm-up.** S8 homework: what did JSON do to `True`? What dirty value did his own CSV throw?
- **0:05–0:30 — Core concept.** Why regex for a researcher (Connection Map #9); raw strings; the survival tokens; `.` matches any char; the four functions; capture groups.
- **0:30–0:42 — Live I.** Validate an email; extract dept+number; collapse whitespace; count hashtags; one case where `.split()` wins.
- **0:42–1:00 — Practice I.** Email validator, extract codes, hashtag count, the "Last, First" flip, the judgment call.
- **1:00–1:08 — Break.**
- **1:08–1:30 — Deeper concept.** Flags (`IGNORECASE`, `MULTILINE`); `re.compile` (name your patterns); `re.VERBOSE` (patterns with comments — reviewable!); named + non-capturing groups; greedy vs lazy (`.+` vs `.+?`); `re.sub` with a **function** (the anonymizer — a closure, S5 pays off); regex + files (S8 bridge).
- **1:30–1:40 — Live II.** Demo deeper sections: flag counts, greedy-vs-lazy brackets, the anonymizer.
- **1:40–1:53 — Practice II.** `VERBOSE` email; the anonymizer; the two-format date harvest; the flag count.
- **1:53–2:00 — Recap + quiz + homework.** `r"..."` always; guard `None`; lazy for bracketed things; compile what you reuse. Quiz S9.

Transitions

- Concept→live: *"Four functions cover almost everything. Every pattern is a raw string, no exceptions."*
- Break→deeper: *"You can write a pattern. Now write patterns a colleague can REVIEW — named, commented, compiled — and do the one trick plain sub can't: computed replacements."*
- Close: *"You can clean any string. Final session of new material: organizing your code so it's reusable."*

Predicted misconceptions

- Forgets raw strings; calls `.group()` on `None`.
- Reaches for greedy `.+` inside brackets/quotes — show the runaway match.
- Expects regex to understand meaning — it matches *form*.

Socratic prompts

- "A theme-mention count: form match or meaning match? What can regex actually promise?"
- "Why did `\\[(.+)\\]` swallow both course codes? What does the `?` change?"
- "Same participant name must always map to the same ID. Where does that memory live?" (→ the closure)

Cut line: drop `MULTILINE` and non-capturing groups; keep `VERBOSE`, greedy-vs-lazy, and the anonymizer. **Homework:** pattern drill · messy-name cleanup · domain harvest.

SESSION 10 — Modules, OOP & the Pythonic Toolkit

Covers: modules & the `__main__` guard, a class with `__init__` / `self` / `__str__`, a validating `@property`, inheritance, generators, `map` / `filter`, walrus · **deeper:** `__repr__` / `__eq__` / `__lt__`, dataclasses with `default_factory`, `@classmethod`, composition vs inheritance, class vs instance attributes, generator pipelines, project layout. **Pre-flight:** `examples/session-10/`; `grades.py` ready to import; the `Student` class and generator-exhaustion demo staged.

The clock (120 min)

- **0:00–0:05 — Warm-up.** S9 homework: run one of his patterns against an *invalid* example — did `fullmatch` catch it?
- **0:05–0:30 — Core concept.** Modules & the `__main__` guard; the `Student` class (`__init__`, `self`, `__str__`, validating `@property` — Connection Map #10); inheritance with `super()`.
- **0:30–0:42 — Live I.** Build the validating `Student`; `ana.gpa = 5.0` raises; the toolkit: comprehension, `map` / `filter`, generator exhaustion, walrus.
- **0:42–1:00 — Practice I.** Import from `grades.py`; the validating `Student`; `GradStudent(super())`; toolkit drills.
- **1:00–1:08 — Break.**
- **1:08–1:30 — Deeper concept.** `__repr__` / `__eq__` / `__lt__` → `sorted()` just works (and `__add__` — operator overloading in one line); dataclasses round 2 (`default_factory` = the S5 rule in class form; `frozen=True`); `@classmethod` alternate constructors (`from_row`: file format at the edge, objects inside — S8 bridge); composition over inheritance; class vs instance attributes (the trap → the law); generator **pipelines** (lazy stages, constant memory); from script to project (folder layout, the `def main()` convention, one module per concern).
- **1:30–1:40 — Live II.** Demo deeper sections: sortable `Score`s, `default_factory`, `from_row`, the two-stage pipeline.
- **1:40–1:53 — Practice II.** Sortable `Student`; the dataclass `Course`; `from_row`; the pipeline.
- **1:53–2:00 — Recap + quiz + homework + course wrap.** Dunders make objects native; mutable data in `__init__` / `default_factory`, always; compose first. Quiz S10. Preview S11: *"Next time, a library does your Session 8 in one line."*

Transitions

- Open (the pair's thread): *"You can clean any string. Last step: organize code so it's reusable — functions into modules, data-plus-rules into classes."*
- Break→deeper: *"You've built a class that defends itself. Now make your objects feel NATIVE — printable, comparable, sortable — and wire generators into a pipeline that could eat a million rows."*
- Close: *"That's the toolbox — every tool earned its place. Next: the libraries researchers build from these same tools."*

Predicted misconceptions

- `self` looks magical; generators exhaust.
- Puts `courses: list = []` in a dataclass — the S5 bug reborn; `default_factory` is the cure.

- Inherits when he should compose — "is-a or has-a?" out loud.

Socratic prompts

- "What must `sorted` be able to ask two Students, for no `key=` to be needed?"
- "Where should `roster = []` live so two Cohorts don't share students? Why?"
- "When does any work happen in `in_range(to_ints(row))` ? What forces it?"

Cut line: drop `frozen=True`, `@staticmethod`, and the project-layout slide; keep dunder, `default_factory`, `from_row`, and the pipeline. **Homework:** `Student` with computed GPA · `Cohort` class · Pythonic rewrite.

SESSION 11 — NumPy & pandas for Research Data

Covers: NumPy arrays, vectorized math, boolean masks, `axis`; pandas `read_csv`, inspection, `loc / iloc`, filtering, new columns, `groupby` + named aggregation · **deeper:** views vs copies, broadcasting, seeded `default_rng`, missing data, `transform`, `melt / pivot_table / crosstab`, `merge`, dates and `resample`, method chaining, the single-`.loc` rule. **Pre-flight:** `examples/session-11/` with `students.csv` + `survey.csv` (the Session 8 files); `import numpy, pandas` works (or use the in-browser notebook, which has both); his Session 8 `demo.py` open side by side for the before/after.

The clock (120 min)

- **0:00–0:05 — Warm-up.** S10 homework: one Pythonic rewrite. Then: "How many lines was your Session 8 by-major mean?"
- **0:05–0:30 — Core concept.** Why arrays (one type, one block, C speed); list `* 2` vs array `* 2`; vectorization; boolean masks and `&`; 2D indexing and `axis`; Series vs DataFrame; `read_csv` infers types.
- **0:30–0:42 — Live I.** Session 8 redone: class mean, curve, pass count in NumPy; load `students.csv`, `info()`, filter, `pd.cut`, `groupby("major").agg(...)`.
- **0:42–1:00 — Practice I.** Vectorize Session 8; the quiz grid + `axis`; load & inspect; filter + grade column; the descriptives table.
- **1:00–1:08 — Break.**
- **1:08–1:30 — Deeper concept.** Views vs copies (S2 aliasing returns); broadcasting (weighted totals, column z-scores); `default_rng(seed)` simulations; missing data (`na_values`, `NaN` → `float64`, drop vs impute as a *methods* decision); `transform`; wide ↔ long (`melt`, `crosstab`, `pivot_table`); `merge` with `validate / indicator`; `.dt`, `resample`, `rolling`; method chaining; the single-`.loc` rule.
- **1:30–1:40 — Live II.** Demo deeper sections: the view surprise, z-scores by broadcasting, the survey melted long, the advisors merge that "loses" Sociology.
- **1:40–1:53 — Practice II.** Broadcast z-scores; clean the survey; melt + crosstab; above-major-mean via `transform`; merge advisors; weekly resample.
- **1:53–2:00 — Recap + quiz + homework.** `&` not `and`; slices are views; `NaN` makes floats; one `.loc` to assign. Quiz S11. Frame the capstone: *"pandas is allowed."*

Transitions

- Open: *"In Session 8 you wrote the loop. Today a library writes it for you — and because you wrote it once, you'll know what it's doing."*
- Break→deeper: *"Clean data is the exception. Second hour: the messy parts — missing values, wide survey exports, joining tables, dates."*
- Close: *"You can now do in six lines what took sixty. Next time you drive: the capstone."*

Predicted misconceptions

- Writes `and` between masks; forgets the parentheses around each condition.

- Expects `arr[:2]` to copy (it did for lists).
- Mixes up `axis=0` and `axis=1` — have him say which dimension disappears.
- Reaches for `.apply` or a `for` loop over rows out of habit.
- Writes `df[mask]["col"] = value` and is puzzled nothing changed.

Socratic prompts

- "Your Likert column now prints `4.0`. What single cell caused that, and why?"
- "Drop the incomplete respondents or impute? Which one would a reviewer ask you to justify?"
- "The merge has fewer advisors than students. How would you *prove* which rows didn't match?"

Cut line: drop `rolling`, `pivot_table`, and the dice simulation (leave the slides as reference); keep masks, `axis`, `groupby`, missing data, `melt`, `merge`, and the `.loc` rule. **Homework:** Session 8 survey summary in six lines · seeded class-mean simulation · your own `groupby` table + cleaning log.

SESSION 12 (Optional) — Capstone

Role shift: you stop teaching and start *coaching*. He drives; you ask questions and unblock. Plan ~2 hours.

- **0:00–0:15 — Brief & plan.** He restates the goal and sketches the steps aloud (pseudocode). You only check the plan is sound.
- **0:15–1:05 — Build.** He codes the Gradebook & Survey Analyzer (`assessments/capstone-project.md`). Intervene only when stuck >3 min; prefer a question over an answer.
- **1:05–1:13 — Break.**
- **1:13–1:45 — Build, continued.** Finish the report CSV, then one stretch goal (a `Student` class, a regex validation, or the recursive nested-data total).
- **1:45–1:55 — Review.** Walk his code for the course's traps (identity, aliasing, mutable defaults, bare `except:`). Praise readability.
- **1:55–2:00 — Debrief & next steps.** Point to pandas/visualization as the genuine next course.

Coaching prompts: "What's your data structure?" · "What happens if that cell is blank?" · "Is that comparing value or identity?" · "Could that be one comprehension?"

Instructor's running checklist (use across all sessions)

- [] Timer visible; both practice blocks got their minutes; the break was honored.
- [] The Going-deeper block taught NEW material — it never became overflow practice time.
- [] Warm-up pulled from the homework and the misconceptions log — not improvised.
- [] Pair through-lines named out loud (S2, S4, S6, S8, S10 openings).
- [] Every trap demoed via **predict-then-run**, not narration.
- [] Each new concept hooked to the **Connection Map** before syntax.
- [] Student typed everything himself; you talked less than half the session.
- [] End-of-session quiz given; homework assigned by name; capstone kept in view as the destination.

Appendix A — Session-by-Session Coverage Checklist

Session-by-Session Coverage Checklist

The teacher's tick-list: every point each session must land, split by half. Print it, keep it next to the clock, and check items off as you go. A session is "covered" when every unticked box has been *deliberately* cut (see each session's cut line in the playbook), not forgotten. Codes: **C** = core hour, **D** = going deeper, **T** = trap to demo predict-then-run.

Session 1 — Running Python, Variables & Types

C — Core hour

- [] REPL (`python3` , `>>>`) vs script (`python3 file.py`); when to use each
- [] Variables are **labels on objects**, not boxes; `=` is an action, so `n = n + 1` is legal
- [] The five core types: `int` , `float` , `str` , `bool` , `None` ; `type(x)` to inspect
- [] `input()` **always returns str** → `"5" + "3" == "53"` ; convert immediately
- [] `print()` with commas (space-joined) vs `+` (same types only)
- [] f-strings: `{x}` , `:.1f` , `:` , `:.1%`
- [] Casting: `int()` , `float()` , `str()` ; `int(3.9)` **truncates**; `round()` rounds
- [] Reading a traceback: **last line first** — exception name + offending value

D — Going deeper

- [] Full operator set: `+` `-` `*` `/` `//` `%` `**` ; precedence follows math; parenthesize when unsure
- [] Augmented assignment: `+=` `-=` `*=`
- [] `%` in practice: even/odd, rotating into k groups, `// + %` = whole units + remainder (130 min → 2 h 10 min)
- [] String methods: `.strip()` `.title()` `.lower()` `.count()` `.startswith()` , `in` , `len()` ;
chaining
- [] Strings are immutable — methods **return new strings**; repetition `"ab" * 3` , `"=" * 40` dividers
- [] f-strings round 2: alignment `:<10` `:>8` , `{x=}` debugging form
- [] `print()` fine-tuned: `sep=` , `end=` ; escape sequences `\n` , `\"` ; quote alternation
- [] `import math` : `sqrt` , `floor` , `ceil` , `pi` — "borrowing a toolbox" (full story S8)
- [] Self-help: `help(round)` , `dir(str)` , REPL `_`
- [] Naming: snake_case, meaningful names, `UPPER` constants, comments say *why*, pseudocode-first habit

T — Traps (4): `"5" + "3"` · `"Age: " + 21` → `TypeError` · `int(3.9)` → `3` · `.upper()` doesn't mutate

Practice: C → GPA reporter, age bucket, string-concat trap; D → aligned gradebook lines, name normalizer, `%`-groups **Homework:** unit converter · traceback drill · type-prediction table

Session 2 — The Dynamic-Typing Traps

C — Core hour

- `[] == (value)` vs `is (identity)`; `is` reserved for `None / True / False`
- `[] b = a` does not copy — mutation shows through both names; copy via `list(a) / a[:]`
- `[] bool < int: True == 1, 5 + True == 6, sum(flags)` counts Trues
- `[] isinstance(True, int)` True but `type(True) is int` False — subclass awareness
- `[] /` always float; `//` floors toward $-\infty$ (`-7 // 2 == -4`)
- `[]` Floats are binary: `0.1 + 0.2 != 0.3`; **never == floats** → `math.isclose`
- `[]` Cross-type: `5 == "5" → False` (no error); `5 > "5" → TypeError`
- `[]` List ≠ tuple even with equal contents; sequences compare element-by-element
- `[]` Truthiness: `falsy = 0 0.0 "" [] {} set() None`; `"0" / "False" / [0]` are truthy; `if scores:` idiom
- `[]` The ~18-trap gauntlet, every prediction **out loud**

D — Going deeper

- `[]` `None` = absence; `is None` test; no- return functions give `None`; `None` vs `""` vs `0` (three "nothings")
- `[]` The conversion matrix: `int("4.2")` fails, `int(float("4.2"))` two-step, `str()` takes anything; **convert at the edge, once**
- `[]` `nan: nan != nan, math.isnan`; `inf` comparisons; where NaN appears in real pipelines
- `[]` Mutable vs immutable table; `id()` is what `is` compares; why immutables are safe to share
- `[]` `Decimal("0.1") + Decimal("0.2") == Decimal("0.3")`; build **from strings**; when floats aren't acceptable; `Fraction` exists
- `[]` Chained equality `a == b == c, 1 <= x <= 5`; don't chain `is`

T — Traps (18): `is` on equal lists · alias append · `True == 1` · `5 + True` · `sum(bools)` · `0.1+0.2` · `== 0.3` · `nan == nan` · `7/2` · `-7//2` · `"5" == 5` · `5 > "5"` · list vs tuple · `type(True) is int` · `isinstance(True, int)` · `bool("0")` · `bool([])` · small-int cache 257

Practice: C → paper gauntlet, `clean_score()` (rejects `bool` !); D → conversion predictions, `Decimal` re-run, `describe(x)` **Homework:** trap journal (5 traps, own words) · `approx_equal` · `is_missing`

Session 3 — Control Flow: Conditionals & Loops

C — Core hour

- [] `if/elif/else` ; indentation defines blocks; first true branch wins
- [] **if vs elif** : stacked `if` s are independent questions and ALL fire (double-label bug); `elif` = one ladder
- [] Comparison operators; **chained comparisons** `0 <= score <= 100`
- [] `and/or/not` ; short-circuit; `and / or` return an **operand** (`5 and 0` → `0`; `""` or `"N/A"` idiom)
- [] Never `if x == True` — just `if x:`
- [] `while` (+ Ctrl+C for infinite loops); `for ... in ; range` **excludes the stop**
- [] `for _ in range(3)` — the underscore convention for unused values
- [] `break / continue`
- [] `enumerate / zip` instead of `range(len(...))`
- [] The `while True: ... break` validation loop

D — Going deeper

- [] The ternary `x if cond else y` — tiny choices only
- [] **Return the test itself**: `return score >= 60`, never `if ...: return True`; call helpers bare (`if is_passing(s):`)
- [] `match/case` : literals, `|` alternatives, `_` default; its niche = one value vs literals
- [] `for/else` — search without a `found` flag (else = no break)
- [] Nested loops; `break` exits inner only → function + `return` escape
- [] The four named patterns: accumulator, counter, best-so-far, sentinel

T — Traps (3): `""` or `'Anonymous'` · `range(1,5)` · `all(gen)` with one failure

Practice: C → grade-band classifier (test 89.999/90/90.001!), logic drill, zip pass/fail, validation loop, mutate-while-iterating, double-label bug; D → match/case rewrite, for/else search, best+worst one pass, return-the-test rewrite **Homework**: attendance labeler · number-guessing game · leap-year checker

Session 4 — Data Structures

C — Core hour

- [] The four containers: list (mutable/ordered), tuple (immutable record), dict (key→value), set (unique)
- [] Indexing, negative indices, **slicing** (`[1:3]` , `[ :2]` , `[::-1]`); stop excluded
- [] `.append()` , `.sort()` mutates in place and **returns None**; `sorted()` returns new
- [] Dict access, `.get(k, default)` , add/update, `.items()`
- [] **A list of dicts is a dataset** — rows as dicts, keys as variables
- [] Sets: dedupe, fast membership
- [] Comprehensions: list, filtered, dict, set — "expr, for each, if"
- [] `sorted(key=lambda ...)` , `reverse=True`
- [] **Aliasing**: `b = a` shares; `.copy()` ; `[[0]*3]*3` shares one row

D — Going deeper

- [] Unpacking: `name, score = pair` , `head, *rest` , swap; zip-loops ARE unpacking
- [] Dict power methods: `.setdefault` , `.pop(k, default)` , `d1 | d2` merge, `.keys/.values/.items`
- [] `Counter` : the tally loop in one line; `.most_common(n)`
- [] `defaultdict(list)` grouping
- [] Set algebra: `& | - ^` , subset `<=` ; ordered dedupe `dict.fromkeys`
- [] Multi-key sorts: tuple keys `(-score, name)` ; sort stability
- [] `copy` vs `deepcopy` — nested + mutating inners → `deepcopy`

T — Traps (3): `[[0]*3]*3` grid · `.get()` missing key · `[::-1]` reversed copy

Practice: C → rank, dict comprehension, group pass/fail, dedupe, aliasing reproduce+fix; D → Counter + most_common, defaultdict grouping, multi-key sort, ordered dedupe **Homework**: gradebook dict drill · frequency counter · fix-the-grid

Session 5 — Functions, Scope & Reusability

C — Core hour

- [] `def`, calling, docstrings; a function = a formula/coding scheme (reproducibility)
- [] `return` vs `print` — `x = show(...)` is `None`
- [] Parameters: positional, keyword, defaults; **defaults must be immutable**
- [] `*args` (→ tuple), `**kwargs` (→ dict); unpacking `f(*lst)`, `f(**dct)`
- [] **The mutable-default bug**, shown live twice; fix = `None` + create inside
- [] Scope LEGB; assigning makes a name local → `UnboundLocalError`; avoid `global`, return instead
- [] Type hints document, are **not enforced** (`mypy`); docstring field conventions (`:param: ...`) + Sphinx pointer

D — Going deeper

- [] Functions are objects: assign, pass, dict dispatch — `key=` demystified
- [] `lambda`: one expression only; name anything bigger
- [] Closures: `make_curver(bonus)` factory; capture-by-variable (the lambda-in-loop trap)
- [] A first decorator: `@announce`; write `f = announce(f)` long-hand FIRST; where he'll meet them (`@property`, `@cache`, `@pytest.mark`)
- [] Keyword-only args (`*`, `)` for readable call sites
- [] Doctests: `>>>` examples in docstrings; `python -m doctest`

T — Traps (3): mutable default persists · print-not-return `is None` · closure late binding

Practice: C → grade-functions library, mutable-default reproduce+fix, `summary(*scores)`; D → `apply_to_all`, curve factory, `@announce`, doctest **Homework:** tiny stats library · `mean_ignoring_none` · scope prediction

Session 6 — Recursion & Recursive Thinking

C — Core hour

- [] The two parts: **base case** (stops) + **recursive case** (moves toward it); say the base case aloud first
- [] Trace the call stack building and unwinding (`orderings(3)`)
- [] Each pending call = a stack frame
- [] Recursion vs iteration; loops win for flat sequences
- [] **Nested data is the payoff**: `deep_sum` over lists/dicts/JSON
- [] `RecursionError` ; no tail-call optimization; limit ≈ 1000
- [] Must **return** the recursive call — else silent `None`

D — Going deeper

- [] Memoization: naive `fib` re-solves subproblems; `@functools.cache` (decorator payoff); when caching helps (repeated subproblems only)
- [] Binary search: halve a **sorted** roster; $\log_2$ intuition ($1,000 \rightarrow \sim 10$ looks)
- [] Tree-shaped data: org chart `count_people` ; function shape mirrors data shape
- [] Explicit-stack conversion: `flatten_iter` with a to-visit list — no frames, no limit
- [] The decision checklist: flat $\rightarrow$ loop · nested $\rightarrow$ recursion · repeated $\rightarrow$ cache · deep $\rightarrow$ stack

T — Traps (3): no base case $\rightarrow$ `RecursionError` · missing `return` $\rightarrow$ `None` · `sys.getrecursionlimit() = 1000`

Practice: C $\rightarrow$ recursive `total`, `reverse`, `flatten`, `depth`, two trap-fixes; D $\rightarrow$ binary search + step counter, org-tree count + deepest, `flatten` without recursion **Homework**: `sum_digits` · `deep_count` · loop-vs-recursion paragraph

Session 7 — Exceptions & Defensive Code

C — Core hour

- [] Syntax errors vs runtime exceptions
- [] `try/except`; **keep the try block small**; `except ValueError: pass` = visible deliberate skip (log if in doubt)
- [] The common types on sight: `ValueError`, `TypeError`, `KeyError`, `IndexError`, `ZeroDivisionError`, `FileNotFoundError`
- [] `try/except/else/finally` — `else` = no exception; `finally` = always
- [] `raise ValueError(...)` deliberately; caller decides
- [] EAFP vs LBYL — both valid; EAFP is the Pythonic default
- [] `assert` = developer check, disableable, **never input validation**
- [] A first pytest test: `test_` naming, `assert`, `pytest.raises`

D — Going deeper

- [] The exception family tree; handlers tried top-down → **specific before broad**; catch a tuple; `except Exception` only to log-and-stop
- [] Custom exceptions that carry data (`SurveyError` with `.value`)
- [] `raise ... from` — domain error on top, real cause beneath
- [] `finally` → the cleanup mindset; `with` is this packaged (S8 bridge)
- [] `logging` vs `print`: output vs diary; levels; `basicConfig` format
- [] pytest round 2: `@pytest.mark.parametrize`; arrange-act-assert; test the **edges**; print-only functions aren't testable — return values

T — Traps (4): `int("3.0") · round(2.5)` banker's · `0.1+0.2+0.3 == 0.6` · `int("1_000")`

Practice: C → clean the raw list + rejection log, add a `raise`, one pytest test; D → fix except-order bug, `SurveyError` + `raise from`, logging conversion, parametrize **Homework:** `ask_int` · three pytest cases · error triage

Session 8 — Files, Libraries & Research Data

C — Core hour

- [] `open / with` — auto-close even on crash
- [] Modes: `r` / `w` **truncates immediately!** / `a` append (`f.write` , `\n` is your job) / binary `rb` / `wb` exist
- [] Reading: `.read()` , line-by-line iteration, `.rstrip()` ; file objects exhaust after one pass
- [] CSV in: `csv.DictReader` , rows as dicts, `newline=""` ; **values are strings — convert!**
- [] Why `csv` exists (fields containing commas beat `.split(",")`)
- [] CSV out: `DictWriter` , `writeheader` , `writerow`
- [] Researcher's stdlib: `statistics` (mean/median/stddev), `random` , `datetime` , `pathlib`
- [] `pip install` + PyPI; `cowsay` (installing is trivial)

D — Going deeper

- [] `pathlib` tour: `/` joins, `.exists` , `.stat().st_size` , `.glob / .rglob` , `mkdir(exist_ok=True)` , `read_text` / `write_text`
- [] Encodings: say `encoding="utf-8"` ; Excel's BOM → `utf-8-sig` ; mojibake symptom
- [] `csv.reader` (lists) vs `DictReader` (dicts); `writerows` ; `delimiter=";"`
- [] JSON: nested data, `dumps(indent=2)` / `loads` , `True` → `true` , `None` → `null`
- [] APIs taste: `requests.get(url, params=...).json()` — fetch a URL, parse the JSON
- [] `datetime` : `strptime` (parse) vs `strftime` (format); date arithmetic `.days` ; `fromisoformat`
- [] `random` with `seed()` → **reproducible** sampling; `sample` vs `choice` ; `shuffle`
- [] Scripts with arguments: `sys.argv` (list of strings, `[0]` = script), guard `len()` , `sys.exit("usage")` , slice `[1:]` ; `argparse` pointer
- [] Binary in practice: Pillow `Image.open` , animated GIF (`save_all` , `append_images` , `duration` , `loop`)
- [] pandas teaser: `read_csv` , `.describe()` , `groupby().agg()` , boolean filter, `to_csv`

T — Traps (3): CSV strings `'91' + 1` · `json.dumps(True)` → `true` · exhausted file cursor

Practice: C → per-item survey means skipping dirty values, `survey_summary.csv` , mean by major; D → `pathlib` inventory, days-between, seeded sample, CSV→JSON, argv-guarded `report.py` , Pillow GIF **Homework:** attendance-report pipeline · JSON round-trip · your own CSV

Session 9 — Regular Expressions & Text Cleaning

C — Core hour

- [] Why regex for a researcher: validate / extract / clean / first-pass qualitative coding; matches **form, not meaning**
- [] **Raw strings always:** `r"..."`
- [] Survival tokens: `.` `\d` `\w` `\s` (+ negations) `+` `*` `?` `{m,n}` `^` `$` `[abc]` `[a-z]` `[^abc]` `(...)` `|`
- [] The `.` trap; escaping literals `\.`
- [] The four functions: `search` / `fullmatch` (validation — anchors both ends) / `findall` / `sub`
- [] Capture groups: `group(0/1/2)`, `m.groups()`; **m may be None — guard before `.group()`**
- [] Cleaning recipes: collapse whitespace, `findall` + `Counter`, group-reorder `r"\2 \1"`
- [] When NOT to use regex: `.split()`, `.strip()`, `.replace()`, `.removeprefix()`

D — Going deeper

- [] Flags: `IGNORECASE`, `MULTILINE` (`^$` per line), `DOTALL`; combining with `|`
- [] `re.compile` — name your patterns, reuse as methods
- [] `re.VERBOSE` — commented, reviewable patterns
- [] Named groups `(?P<dept>...)` + `groupdict`; non-capturing `(?:...)`
- [] Greedy vs lazy: `.+` vs `.+?` (bracketed/quoted content wants lazy)
- [] `re.sub` with a **function** replacement — the anonymizer (closure!)
- [] Regex + files: `findall` over `Path.read_text()` (S8 bridge)

T — Traps (3): greedy `\[(.+)\]` · `re.match` anchors at start · `.` in `grades.csv` matches X

Practice: C → email validator, extract codes, hashtag count, "Last, First" flip, judgment call; D → VERBOSE email, anonymizer, two-format date harvest, flag count **Homework:** pattern drill · messy-name cleanup · domain harvest

Session 10 — Modules, OOP & the Pythonic Toolkit

C — Core hour

- [] Modules: a `.py` file; `import` / `from ... import`; no copy-paste reuse
- [] The `if __name__ == "__main__":` guard — script AND library
- [] A class = an operational definition; `__init__`, `self`, attributes, methods, `__str__`
- [] `@property` + setter validation — the object defends its integrity; `_attr` privacy convention
- [] Inheritance: `GradStudent(Student)`, `super().__init__`, override `__str__`
- [] Toolkit: comprehensions, `map`, `filter`, `enumerate`, `zip`
- [] Generators: `yield`, lazy, **exhaust after one pass**; walrus `:=`

D — Going deeper

- [] `__repr__` (developers) vs `__str__` (people); `__eq__`; `__lt__` → `sorted()` with no key; `__add__` = operator overloading
- [] Dataclasses round 2: defaults, `field(default_factory=list)` (the S5 rule in class form), `frozen=True`
- [] `@classmethod` alternate constructor `from_row` (CSV at the edge, objects inside); `@staticmethod` third flavor
- [] Composition over inheritance: Cohort HAS Students; inherit only for true is-a
- [] Class vs instance attributes — mutable data in `__init__`, always; object state beats `global`
- [] Generator pipelines: chained stages, lazy, constant memory; the `iter()/next()` protocol underneath
- [] From script to project: folder layout, one module per concern, `def main()` convention, `tests/` (+ `__init__.py` note)

T — Traps (3): generator exhausts · dataclass `__eq__` · shared class-variable list

Practice: C → import from `grades.py`, validating `Student`, `GradStudent`, toolkit drills; D → sortable `Student`, dataclass `Course`, `from_row`, two-stage pipeline **Homework:** `Student` with computed GPA · `Cohort` class · Pythonic rewrite

Session 11 — NumPy & pandas for Research Data

C — Core hour

- [] Why now: Session 8's by-hand loops are what these libraries do underneath
- [] `np.array`: one dtype, `shape`; `zeros` / `arange` / `linspace`; list `* 2` repeats vs array `* 2` multiplies
- [] Vectorized math and whole-array stats (`mean`, `std`, `np.sqrt`); int arrays truncate assigned floats
- [] Boolean masks: `arr[arr >= 60]`, `.sum()` counts True, `&` / `|` / `~` with parentheses, `np.where`
- [] 2D indexing `[r, c]`, `[:, c]`; `axis=0` vs `axis=1` ("the axis that disappears")
- [] `pd.read_csv` infers types; `head` / `shape` / `info` / `describe`
- [] Selecting: `df["col"]`, `df[...]`, `loc` (labels, end included) vs `iloc` (positions, end excluded)
- [] Filtering with masks, `isin`, `query`
- [] New columns, `pd.cut`, z-score column; `sort_values`, `value_counts`; vectorize before `.apply`
- [] `groupby` + named aggregation = descriptives by condition

D — Going deeper

- [] Views vs copies: array slices are views, `.copy()`; mask/list indexing copies
- [] Broadcasting rule (compare shapes from the right; size 1 stretches): weighted totals, column z-scores
- [] `np.random.default_rng(seed)`: `integers`, `normal`, `choice`; a two-line simulation
- [] Missing data: `na_values`, `isna().sum()`, pandas skips NaN, `dropna` / `fillna`; NaN → float64; report the decision
- [] `transform` vs `agg` (row-aligned group stats)
- [] Reshape: `melt` (wide → long), `pivot_table`, `crosstab`
- [] `merge`: `how=`, `validate=`, `indicator=True`
- [] Dates and text: `to_datetime` / `parse_dates`, `.dt`, `resample`, `rolling`; `.str`
- [] Method chaining; one `.loc[rows, col]` for assignment (chained assignment edits a copy); `to_csv(index=False)`

T — Traps (6): list `* 2` · `and` on arrays · array slice is a view · NaN makes float64 · `loc` slice includes end · index alignment → NaN

Practice: C → vectorize Session 8, quiz grid + `axis`, load & inspect, filter + `pd.cut`, `groupby` table; D → broadcast z-scores, view surprise, clean the survey, melt + `crosstab`, `transform`, merge advisors, weekly resample **Homework:** Session 8 survey summary in six lines · seeded class-mean simulation · your own `groupby` table + cleaning log

Session 12 (Optional) — Capstone

- [] Brief restated by the student in pseudocode before any code
- [] Reads both CSVs; cleans & validates (skip/flag dirty, range checks)
- [] Computes: class mean/median/stdev, mean by major, per-item survey means + n
- [] Flags at-risk students (< 60)
- [] Writes `report.csv`
- [] Organized: helper module or class, docstrings + hints, `main()` + guard
- [] Review pass hits the course traps: identity, aliasing, mutable defaults, bare except
- [] One stretch goal attempted (class / regex / recursive total / argparse)
- [] Debrief: pandas & visualization named as the next course

Appendix B — Master Course Outline

Master Course Outline — Learn Python

Single source of truth. The student and teacher syllabi both derive from this. **10 core two-hour sessions** + 1 optional capstone session (~20 hours core, ~22 with the capstone). Every session pairs a **core** hour with a **Going deeper** hour of new material, and assigns **homework (~30–45 min)** that does *not* count toward class time.

How the topics are sequenced

A typical intro-Python course teaches roughly: functions/variables → conditionals → loops → exceptions → libraries → unit tests → file I/O → regular expressions → OOP → assorted "power-tools."

We **re-sequence** for a fast adult learner, and we front-load the dynamic-typing traps — they're the whole reason this course exists. The 10 sessions run in five natural pairs, each pair sharing a through-line (name it when you cross between them):

1. **Types (S1) → the type traps (S2)**. You can't reason about the traps until you know the types — so they lead, back to back, on days one and two.
2. **Control flow (S3) → data structures (S4)**. Loops are most useful over the containers worth looping; a list of dicts is just a dataset.
3. **Functions (S5) → recursion (S6)**. Recursion is a function that calls itself, so it lands while function mechanics are fresh — and nested data (from S4) is the payoff.
4. **Exceptions (S7) → files & research data (S8)**. Surviving one dirty value scales straight into cleaning a whole CSV of them.
5. **Regex (S9) → modules & OOP (S10)**. The two "finishing" skills: clean any text, then organize code into modules and a small class.

The power-tools (comprehensions, generators, `*args`, type hints, the walrus) are folded into the sessions where they naturally belong, not saved for the end.

Design rules (from the e-learning pipeline)

- **Every session = two hours = one topic, twice as deep:** warm-up → core concept → live example → practice → **break** → going-deeper concept → live example → practice → recap + quiz. The second hour is **new material** (never overflow practice); the two practice blocks total ~30 min of protected hands-on time.
 - Every session assigns **homework** (in `examples/session-NN/practice.md` → *Homework*): ~30–45 min outside class, solutions included, reviewed in the next session's warm-up. The practice file's **In class — going deeper** block belongs to the second hour.
 - Every session has 3–5 learning objectives across its core and going-deeper halves; every objective is testable in that session's quiz.
 - Every abstract idea ships with a runnable, education-flavored example.
 - Difficulty rises monotonically; nothing is used before it's introduced (except clearly-flagged teasers).
-

Session 1 — Running Python, Variables & Types

Objectives

1. Run Python interactively (REPL) and as a `.py` script; read a traceback without panic (last line first).
2. Use the core types (`int` , `float` , `str` , `bool` , `None`); check them with `type()` .
3. Do I/O with `input()` / `print()` , f-strings, and `int()` / `float()` / `str()` casting — and remember `input()` is *always* a `str` .

Going deeper (second hour): the full operator set (`//` `%` `**`), string methods, aligned f-strings, `print(sep=, end=)` & escape sequences, `import math` , `help()` / `dir()` , naming conventions. **Trap focus:** `input()` returns text (`"5" + "3"` is `"53"`); `int(3.9)` truncates; `print(a, b)` vs `print(a + b)` . **Homework:** unit converter; traceback drill; type-prediction table.

Session 2 — The Dynamic-Typing Traps (*the core of the course*)

Objectives

1. Distinguish **value equality** (`==`) from **identity** (`is`); keep `is` for `None`.
2. Predict `bool < int` (`True == 1`, summing flags), int/float mixing, and **float precision** (`0.1 + 0.2`); compare floats with `math.isclose`.
3. Handle cross-type comparison (`5 == "5"` is `False`, `5 > "5"` raises); check types with `isinstance()` vs `type()`; reason about truthiness.

Why it leads: front-loading the traps — right after the types they concern — means every later session can assume the fluency. **Going deeper (second hour):** `None` semantics, the conversion matrix, `nan / inf`, mutable vs immutable + `id()`, exact arithmetic with `Decimal`. **Trap focus:** the ~18-trap predict-then-run gauntlet *is* the session. **Homework:** trap journal (5 traps in your own words); `approx_equal`; `is_missing`.

Session 3 — Control Flow: Conditionals & Loops

Objectives

1. Write `if / elif / else` with comparison & logical operators and **chained comparisons**; use `and / or / not` correctly (short-circuit, operand-return); avoid `if x == True`.
2. Write `for / while` loops; control them with `break / continue`; mind the `range` off-by-one; run the `while True:` validation loop.
3. Iterate Pythonically with `enumerate / zip` instead of `range(len(...))`.

Going deeper (second hour): the ternary, the boolean-return idiom, `match/case`, `for/else`, nested loops, the named loop patterns (accumulator, counter, best-so-far, sentinel). **Trap focus:** `if x == True`; stacked `if`s that all fire where an `elif` ladder was meant; `range(1,5)` excludes 5; mutating a list while iterating it; `range(len(...))`. **Homework:** attendance labeler (boundary testing); number-guessing game; leap-year checker.

Session 4 — Data Structures

Objectives

1. Choose between `list` / `tuple` / `dict` / `set` ; index, slice, and nest them — a **list of dicts is a dataset**.
2. Build list/dict **comprehensions**; sort with `sorted(key=...)` .
3. Reason about **mutability & aliasing** (copy vs reference, `[[0]*3]*3` shared rows).

Going deeper (second hour): unpacking, dict power methods, `Counter` / `defaultdict` , full set algebra, multi-key sorting, `copy` vs `deepcopy` . **Trap focus:** aliasing (`b = a` shares the list); `.sort()` returns `None` ; `dict.get()` vs `KeyError` ; the shared-row grid. **Homework:** gradebook dict drill; frequency counter; fix-the-grid.

Session 5 — Functions, Scope & Reusability

Objectives

1. Define functions with positional, keyword, default, `*args`, and `**kwargs` parameters; know `return` vs `print`.
2. Explain scope (LEGB) and dodge the `UnboundLocalError` / `global` trap.
3. Document with docstrings and **type hints** (and know hints aren't enforced).

Going deeper (second hour): functions as values, closures, a first decorator, keyword-only arguments, doctests. **Trap focus:** the **mutable default argument** bug; forgetting to `return`; assigning to a global inside a function. **Homework:** a tiny stats library; `mean_ignoring_none(*values)`; scope-prediction drill.

Session 6 — Recursion & Recursive Thinking

Objectives

1. Write a recursive function with a correct **base case** and **recursive case**; trace the **call stack**.
2. Convert between recursion and iteration; know recursion's cost (`RecursionError` , no tail-call optimization, limit ≈ 1000).
3. Apply recursion to **naturally nested data** (nested lists/dicts/JSON) where one loop is awkward.

Going deeper (second hour): memoization (`@functools.cache`), binary search, tree-shaped data, the explicit-stack escape from the recursion limit. **Trap focus:** an unreachable base case $\rightarrow$ stack overflow; forgetting to `return` the recursive call. **Homework:** `sum_digits` ; `deep_count` on a nested gradebook; loop-vs-recursion paragraph.

Session 7 — Exceptions & Defensive Code

Objectives

1. Handle errors with `try / except / else / finally`; `raise` deliberately; validate messy human/research input the EAFP way.
2. Know the common exception types on sight (`ValueError` , `TypeError` , `KeyError` , `FileNotFoundError` , ...).
3. Use `assert` for developer checks (never input validation) and write a first `pytest` test.

Going deeper (second hour): the exception hierarchy & handler order, custom exceptions that carry data, `raise ... from`, `logging`, parametrized tests. **Trap focus:** bare `except:`; swallowing errors; `assert` ≠ validation. **Homework:** `ask_int` (EAFP validation loop); three more `pytest` cases; error triage.

Session 8 — Files, Libraries & Research Data

Objectives

1. Read/write text with `open / with` and understand file modes (why `"w"` is dangerous).
2. Load and write **CSV** survey/gradebook data with `csv.DictReader / DictWriter`; touch `json`.
3. `import` the researcher's stdlib (`statistics`, `datetime`, `pathlib`), `pip install` a package, and meet `pandas` in a guided teaser.

Going deeper (second hour): `pathlib`, encodings, `csv.reader` vs `DictReader`, JSON for nested data, `datetime`, seeded `random`, scripts with `sys.argv / sys.exit` (+ the `argparse` pointer), `pip` & PyPI (`cowsay`), a `requests` API taste, binary files & animated GIFs with Pillow, the extended `pandas` teaser. **Trap focus:** `"w"` silently overwrites; forgotten `newline=""`; reading a file twice; CSV values are strings. **Homework:** attendance-report pipeline; JSON round-trip; run the pipeline on your own CSV.

Session 9 — Regular Expressions & Text Cleaning

Objectives

1. Write patterns with raw strings and the survival tokens; know when *not* to use regex.
2. Use `re.search` / `fullmatch` / `findall` / `sub` to validate, extract (capture groups), and clean real research text.

Going deeper (second hour): flags, `re.compile`, `re.VERBOSE`, named groups, greedy vs lazy, `re.sub` with a function replacement, regex over files. **Trap focus:** forgetting `r"..."`; `.` matches any char; `re.search` returns `None` — guard before `.group()`. **Homework:** pattern drill (IDs, phones, dates); messy-name cleanup; domain harvest.

Session 10 — Modules, OOP & the Pythonic Toolkit

Objectives

1. Split code into modules and `import` them; understand the `if __name__ == "__main__":` guard.
2. Model a domain entity with a small **class** (`__init__` , `self` , `__str__` , a validating `@property` , brief inheritance with `super()`).
3. Apply the **Pythonic toolkit**: comprehensions, `map` / `filter` , generators/ `yield` , the walrus `:=` .

Going deeper (second hour): `__repr__` / `__eq__` / `__lt__` and `__add__` , dataclasses with `default_factory` , `@classmethod` constructors, composition vs inheritance, generator pipelines, project layout with `def main()` . **Trap focus:** `self` confusion; a generator exhausts after one pass; the shared class variable; over-using a class where a function/dict fits. **Homework:** `Student` with computed GPA; a `Cohort` class; Pythonic rewrite of three loops.

Session 11 — NumPy & pandas for Research Data

Objectives

1. Replace Python loops with **NumPy** arrays: vectorized math, boolean masks, and aggregations along an `axis`.
2. Load, inspect, select (`loc` / `iloc`), filter, and add columns to a **pandas** DataFrame.
3. Produce "descriptives by condition" with `groupby(...).agg(...)` and named aggregation, redoing Session 8's by-hand work in a line each.

Going deeper (second hour): views vs copies, broadcasting (z-scores, weighted totals), seeded `default_rng` simulations, missing data (`na_values` , `isna` , `dropna` / `fillna` , `NaN` → `float64`), `transform` , `melt` / `pivot_table` / `crosstab` , `merge` with `validate` and `indicator` , `.dt` / `.str` , `resample` / `rolling` , method chaining, the single- `.loc` assignment rule, `to_csv` . **Trap focus:** list `*` 2 repeats; `and` on arrays; array slices are views; one `NaN` makes floats; `loc` slices include the end; index alignment produces `NaN`. **Homework:** Session 8's survey summary in six pandas lines; a seeded class-mean simulation; one publishable `groupby` table from your own data plus its cleaning log.

Session 12 (Optional) — Capstone Project (*integrative*)

Objective: independently build one small, end-to-end program on a real-ish education dataset. Default brief: **"Gradebook & Survey Analyzer"** — read a CSV of students + Likert responses, clean and validate it, compute summary statistics, flag at-risk students, and write a report CSV. Alternative briefs are listed in `assessments/capstone-project.md`. Plan ~2 hours.

Coverage check (every core topic lands somewhere)

Topic	Where it lives now
Running Python, variables & types	S1
The dynamic-typing traps (<code>==</code> / <code>is</code> , floats, <code>bool</code> <code>int</code> , <code>isinstance</code>)	S2
Conditionals & loops	S3
Data structures (list/tuple/dict/set)	S4
Functions, scope & reuse	S5
Recursion	S6 (nested data ties back to S4)
Exceptions & unit tests	S7
Files & libraries (CSV, <code>statistics</code> , <code>pandas</code> teaser)	S8
Regular expressions	S9
Modules & OOP	S10
NumPy & pandas (arrays, DataFrames, <code>groupby</code> , reshape, merge)	S11 (teaser in S8)
Power-tools (comprehensions, <code>*args</code> , type hints, generators, <code>map</code> / <code>filter</code> , walrus)	S4, S5, S10

Scaling to the time budget

- **~14–16 hours:** keep every core hour; trim each Going-deeper block to its two or three starred essentials (each session's *cut line* names them).
- **~22 hours:** run S1–S11 as written, two hours each (recommended).
- **~24 hours:** add the S12 capstone.

Appendix C — Connection Map (education-research bridges)

Connection Map — Python $\rightleftharpoons$ Education-Research Background

From the *personalized-syllabus* method: each new programming concept is bridged to something the student already knows from education research, and — crucially — **where the bridge breaks** is named, so the new concept isn't quietly collapsed into the familiar one. Use these as the "hooks" when introducing each topic; they cut learning time dramatically for an expert adult.

1. Variables & assignment (S1)

- **Maps onto:** named quantities in a stats model — `n`, `mean`, `alpha`.
- **Where it breaks:** in math, `x = x + 1` is a contradiction; in Python `=` is an *action* (rebind the name), not an assertion of equality. A variable is a **label stuck on an object**, not a box that holds a value. This single reframing prevents most aliasing confusion later.

2. Types & dynamic typing (S1–S2)

- **Maps onto:** measurement levels — nominal/ordinal/interval/ratio. Python's `str` / `bool` / `int` / `float` are loosely analogous (categorical vs counted vs continuous).
- **Where it breaks:** SPSS/R columns have *one declared type per column*; a Python variable can hold a string now and an int later. The discipline a column gives you for free, you must supply yourself. This is exactly why the comparison traps in S2 exist.

3. `==` vs `is` (S2)

- **Maps onto:** the difference between **two questionnaires with identical answers** (equal in value) and **the same physical respondent** (identical individual). Two students can have the same GPA (`==`) without being the same person (`is`).
- **Where it breaks:** the analogy is exact for objects, but Python adds an implementation wrinkle (small-integer caching) that has no analogue in research — flagged as a "do not rely on this."

4. Boolean as a subclass of int (`True == 1`) (S2)

- **Maps onto:** **dummy coding** — you already encode "treatment = 1, control = 0" and then *sum* the dummies to count cases. Python literally does this: `sum([True, False, True]) == 2`.

- **Where it breaks:** in your data the 1/0 is a convention you imposed; in Python it's baked into the language, so `True + True == 2` works even when you didn't intend arithmetic on a flag.

5. Float precision (`0.1 + 0.2 != 0.3`) (S2)

- **Maps onto: rounding/measurement error.** You already never test two measured scores for exact equality; you use tolerances and report to 2 decimals.
- **Where it breaks:** here the error isn't in the data — it's in how the *computer* stores decimals in binary. Same instinct (`math.isclose` , round for display), new cause.

6. Lists / dicts / DataFrames (S4, S8)

- **Maps onto:** a `list` of `dict`s is a **tidy dataset**: each dict is a *row/respondent*, each key is a *variable/column*. A `dict` alone is one record's variable→value map.
- **Where it breaks:** a spreadsheet enforces rectangularity; a list of dicts does not — rows can have missing or extra keys, which is the source of many bugs (and why we validate in S7).

7. Functions (S5)

- **Maps onto:** a **statistical formula or a coding scheme**: defined once, applied to many cases, same rule every time → reproducibility, the thing you care about most as a researcher.
- **Where it breaks:** functions can carry *hidden state* (mutable defaults, globals) so the "same input → same output" promise can silently fail. That trap (S5) is the whole reason to learn scope.

8. Exceptions & validation (S7)

- **Maps onto: data cleaning** — out-of-range Likert values, blank cells, "N/A" typed into a numeric field. You already have a mental model of dirty data; exceptions are how code reacts to it.
- **Where it breaks:** cleaning in SPSS is a one-time batch pass; in a program you decide *at runtime*, per value, whether to skip, fix, or stop — a more granular control than a recode syntax file.

9. Regular expressions (S9)

- **Maps onto: search-and-filter in a corpus / qualitative coding** — finding every response matching a pattern, extracting IDs, normalizing free-text.
- **Where it breaks:** regex matches *surface form*, not meaning. It's powerful for structure (emails, dates, IDs) and treacherous for semantics — the opposite trade-off from human coding.

10. Classes / OOP (S10)

- **Maps onto:** an **operational definition / construct** — a `Student` class bundles the attributes and behaviors you've decided "count" as a student, like an operationalized variable.

- **Where it breaks:** a construct is a measurement choice; a class is also *behavior* (methods) and *enforced rules* (validation in setters), so it's a construct that can defend its own integrity.

11. Recursion (S6)

- **Maps onto:** a **hierarchy / nested structure** — coding schemes with sub-codes, threaded discussion data, folder trees of data files, nested JSON survey exports. A procedure "defined in terms of itself" mirrors data that contains smaller copies of itself.
 - **Where it breaks:** recursion is elegant only when the structure is genuinely nested and the base case is reachable; for a flat sequence, a loop is clearer, and Python's stack limit (~1000, no tail-call optimization) makes very deep recursion a liability rather than a virtue.
-

Questions to hold while learning (Socratic spine)

These recur across sessions; the teacher should keep returning to them.

1. *Is this a question about value, or about identity?* (drives `==` vs `is`, copy vs alias)
2. *What type is this right now, and who decided that?* (dynamic typing discipline)
3. *Could this input be dirty? What happens if it is?* (defensive mindset)
4. *Is the same rule guaranteed to run the same way every time?* (reproducibility / pure functions)
5. *Is there a more Pythonic, more readable way to say this?* (the Zen of Python's "readability counts")

Appendix D — Per-Session Quizzes & Answer Keys

Per-Session Quizzes (with answer keys)

Each quiz is 5–6 questions (the last one or two from the Going-deeper hour), ~8 minutes, given at the end of the session. Every question maps to a session objective. **Teacher:** ask the student to *predict* code output before they run it — the surprise is the assessment. Answers are at the end of each session block.

Session 1 — Running Python, Variables & Types

1. What type does `input("x: ")` return, always?
2. What does `"5" + "3"` evaluate to? How do you get `8`? And what is `int(3.9)`?
3. What does `f"{0.873:.1%}"` produce?
4. In a traceback, which line do you read first — and what two things does it tell you?
5. What are `17 // 5` and `17 % 5` — and name one practical use of `%`.

Answers: 1. `str`. 2. `"53"`; use `int("5") + int("3")`; `int(3.9)` is `3` (truncates, doesn't round). 3. `"87.3%"`. 4. The **last** line: the exception's name (what kind of problem) and the message (the offending value or detail). 5. `3` and `2`; `%` gives the remainder — even/odd tests, rotating IDs into k groups, splitting minutes into h + min.

Session 2 — The Dynamic-Typing Traps

1. `a = [1,2]; b = [1,2]` — is `a == b` ? Is `a is b` ? Why?
2. What is `True + True` ? Why?
3. Is `0.1 + 0.2 == 0.3` True or False? How *should* you compare them?
4. What does `5 == "5"` give? What about `5 > "5"` ?
5. Which of these are falsy: `"0"` , `""` , `[0]` , `[]` , `None` , `0.0` ?
6. Why does `Decimal("0.1") + Decimal("0.2") == Decimal("0.3")` hold when the float version doesn't — and why build Decimals from strings?

Answers: 1. `==` True (same value), `is` False (different objects in memory).

2. `2` — `bool` is a subclass of `int` (`True` acts as `1`). 3. False (binary float rounding); use `math.isclose(0.1 + 0.2, 0.3)` or `round`. 4. `False` (no error); `5 > "5"` raises `TypeError` — Python can't *order* `int` vs `str`. 5. `""` , `[]` , `None` , `0.0` are falsy; `"0"` and `[0]` are *truthy* (non-empty). 6. `Decimal` stores base-10 digits exactly, so there's no binary rounding; built from a float (`Decimal(0.1)`) it would inherit the float's error.
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Session 3 — Control Flow: Conditionals & Loops

1. Rewrite `if x >= 90 and x < 100:` using a chained comparison.
2. What is `5 and 0`? What is `0 or "hi"`? Why aren't they `True / False`?
3. What does `list(range(1, 5))` produce?
4. What goes wrong when you `remove` items from a list while looping over it — and what's the fix?
5. When does the `else` clause on a `for` loop run?
6. With `score = 95`, three stacked `if score >= 60 / 80 / 90:` prints run — what happens, and what one keyword fixes it?

Answers: 1. `if 90 <= x < 100:`. 2. `0` and `"hi"` — `and / or` return an *operand*, not a bool. 3. `[1, 2, 3, 4]` (5 excluded — off-by-one). 4. Removal shifts the remaining indices, so elements get skipped; build a new list instead (`[x for x in xs if keep(x)]`) or iterate over a copy. 5. Only when the loop finishes **without** `break` — the "searched everything, found nothing" branch. 6. All three labels print (each `if` is an independent question); chaining with `elif` makes one ladder where only the first hit runs.

Session 4 — Data Structures

1. Which are mutable: list, tuple, dict, set?
2. `a = [1,2]; b = a; a.append(3)` — what is `b`? Why?
3. (Practical) Write a one-line dict comprehension mapping each name in `names` to `0`.
4. `xs.sort()` vs `sorted(xs)` — what does each return?
5. You need "have I seen this ID before?" over thousands of IDs — which structure, and why?
6. What does `Counter(['a', 'b', 'a']).most_common(1)` return?

Answers: 1. list, dict, set are mutable; tuple is not. 2. `[1, 2, 3]` — `b` is an alias for the *same* list (assignment doesn't copy). 3. `{n: 0 for n in names}`.

4. `.sort()` sorts in place and returns `None`; `sorted()` returns a *new* sorted list.
 5. A `set` — membership tests are fast and duplicates are impossible by construction.
 6. `[('a', 2)]` — a list of (value, count) pairs, biggest first.
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Session 5 — Functions, Scope & Reusability

1. What's the difference between `return` and `print`?
2. Why is `def f(x, items=[])` dangerous? What's the fix?
3. What does a function with no `return` statement return? Are type hints enforced at runtime?
4. `count = 0` at top level, then inside a function `count = count + 1` — what error, and why?
5. In one sentence: what does a decorator do, and what is `@d` above a `def f` sugar for?

Answers: 1. `return` hands a value to the caller; `print` only displays it.

2. The default list is created once, at `def` time, and persists across calls; use `items=None` then create inside. 3. `None`; and no — hints are documentation (`mypy` checks them optionally). 4. `UnboundLocalError` — the assignment makes `count` local to the function, so the right-hand side reads a local that doesn't exist yet. 5. It wraps a function with extra behavior; `@d` is exactly `f = d(f)`.
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Session 6 — Recursion & Recursive Thinking

1. What two parts must every recursive function have?
2. What error comes from a base case that's never reached, and why doesn't Python just keep going?
3. Why is recursion a natural fit for nested data (a list of lists, or nested JSON)?
4. The recursive case computes `n * f(n - 1)` but has no `return` in front — what does a call produce?
5. Naive recursive `fib(35)` takes seconds but `@functools.cache` makes it instant — what does the cache change?

Answers: 1. A **base case** (stops) and a **recursive case** (calls itself on a smaller input, *toward* the base case). 2. `RecursionError` — Python has no tail-call optimization, so each pending call keeps a stack frame until the limit (~1000).

3. The data is *defined in terms of itself* (a list may contain lists), so a function defined in terms of itself mirrors its shape and reaches every level. 4. `None` — the value is computed and thrown away; the function falls off the end. 5. Each distinct input is computed once and remembered, so the exponential tree of repeated subcalls collapses into ~35 lookups.
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Session 7 — Exceptions & Defensive Code

1. Which exception does `int("N/A")` raise? Wrap `int(value)` so it returns `None` on failure.
2. Why is a bare `except:` dangerous?
3. When should you use `raise` vs `assert`?
4. Name the two validation styles — and give the EAFP version of converting `value` to an int.
5. Why must `except ValueError:` come before `except Exception:`?

Answers: 1. `ValueError`; `try: return int(value) / except (ValueError, TypeError): return None`.

2. It catches *everything* (even Ctrl+C and your own typos) and can hide bugs.
 3. `raise` to validate real/untrusted input; `assert` for developer sanity checks (it can be disabled with `python -O`).
 4. LBYL ("look before you leap": `if value.isdigit():`) vs EAFP ("easier to ask forgiveness"): `try: n = int(value) / except ValueError: ...`
 5. Handlers are tried top-down and `Exception` matches a `ValueError` too — placed first it would swallow everything, leaving the specific handler unreachable.
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Session 8 — Files, Libraries & Research Data

1. What does opening a file in `"w"` mode do to existing contents? Why prefer `with open(...)` ?
2. After `csv.DictReader` , what type is each row?
3. CSV values read from a file are what type — and what must you do with numbers?
4. Your second loop over an open file object sees nothing — why, and what's the fix?
5. What do `strptime` and `strftime` each do — and why seed `random` in an analysis script?
6. Your script runs as `python3 report.py survey.csv` — what is `sys.argv` , and what sits at index 0?

Answers: 1. Truncates it to empty *immediately*; `with` auto-closes the file even if the code crashes. 2. A `dict` keyed by the header row. 3. Strings; convert with `int()` / `float()` . 4. The file cursor is exhausted after one pass — re-open the file (or read it into a list first). 5. `strptime` parses text → datetime; `strftime` formats datetime → text; seeding makes the "random" sample reproducible for your methods section. 6. A plain list of strings, `['report.py', 'survey.csv']` ; index 0 is always the script's own name, so real arguments start at 1 — check `len(sys.argv)` and `sys.exit("usage...")` before trusting them.

Session 9 — Regular Expressions & Text Cleaning

1. In regex, what does `.` match? How do you match a literal dot?
2. Why write regex patterns as raw strings `r"..."` ?
3. What does `re.search` return when there's no match, and what must you do before `.group()` ?
4. `re.search` vs `re.fullmatch` — which one is for validation, and why?
5. What does `re.IGNORECASE` change, and what is `re.VERBOSE` for?

Answers: 1. Any character (except newline); use `\.` for a literal dot. 2. So backslashes aren't treated as Python string escapes. 3. It returns `None` ; check `if m:` before `m.group()` or you'll hit an `AttributeError`. 4. `fullmatch` — it anchors both ends, so the *whole* string must fit the pattern; `search` would accept a valid-looking fragment inside garbage. 5. `IGNORECASE` matches regardless of case; `VERBOSE` lets a pattern span lines with comments so a colleague can actually review it.

Session 10 — Modules, OOP & the Pythonic Toolkit

1. In a class, what is `self`?
2. What happens if you iterate a generator twice?
3. Why doesn't a module's `if __name__ == "__main__":` block run when you `import` it?
4. `class Course: students = []` — what goes wrong when two `Course` objects enroll students, and what's the fix?
5. In a dataclass, why write `courses: list = field(default_factory=list)` instead of `= []`?

Answers: 1. The current instance ("this particular object"). 2. The second pass is empty — a generator is exhausted after one iteration. 3. On import, `__name__` is the module's name, not `"__main__"`, so the block is skipped. 4. `students` is a *class* variable shared by every instance, so both courses see one combined list; give each its own in `__init__`: `self.students = []`. 5. A bare `[]` default would be created once and shared by every instance — the Session 5 mutable-default bug in class form; `default_factory` builds a fresh list per instance.

Session 11 — NumPy & pandas for Research Data

1. `[1, 2] * 2` vs `np.array([1, 2]) * 2` — what does each give, and why?
2. Why does `(scores > 60)` and `(scores < 90)` raise an error on an array, and what do you write instead?
3. For a `(students, quizzes)` array, which gives one mean per student: `axis=0` or `axis=1`?
4. One cell of an integer survey column is blank. What happens to the column's dtype, and what does `.mean()` do with the blank?
5. `groupby(...).agg(...)` vs `groupby(...).transform(...)` — what shape does each return?
6. Why does `df[df["score"] < 60]["grade"] = "D"` leave `df` unchanged, and what's the fix?

Answers: 1. `[1, 2, 1, 2]` (a list repeats) vs `array([2, 4])` (an array does element-wise math). 2. `and` needs a single True/False, but the comparison gives a whole array of them, so NumPy refuses to guess (ValueError); write `(scores > 60) & (scores < 90)`, with the parentheses. 3. `axis=1`: the axis you name is the one that disappears, so collapsing the quiz columns leaves one value per student. 4. It becomes `float64` (NaN is a float); pandas skips NaN, so the mean is over the valid cells only. 5. `agg` returns one row per group; `transform` returns one value per original row, aligned to the index, ready to become a new column. 6. The filter makes a copy and the assignment lands on that copy (pandas warns); use one `.loc`: `df.loc[df["score"] < 60, "grade"] = "D"`.

Scoring guide (formative, not graded)

- **All correct, explained why:** ready for the next session (or the capstone).
- **Right answer, fuzzy why:** re-do that session's traps-and-gotchas rows.
- **Wrong on Session 2 items:** revisit the type traps before continuing — they're load-bearing for everything after.